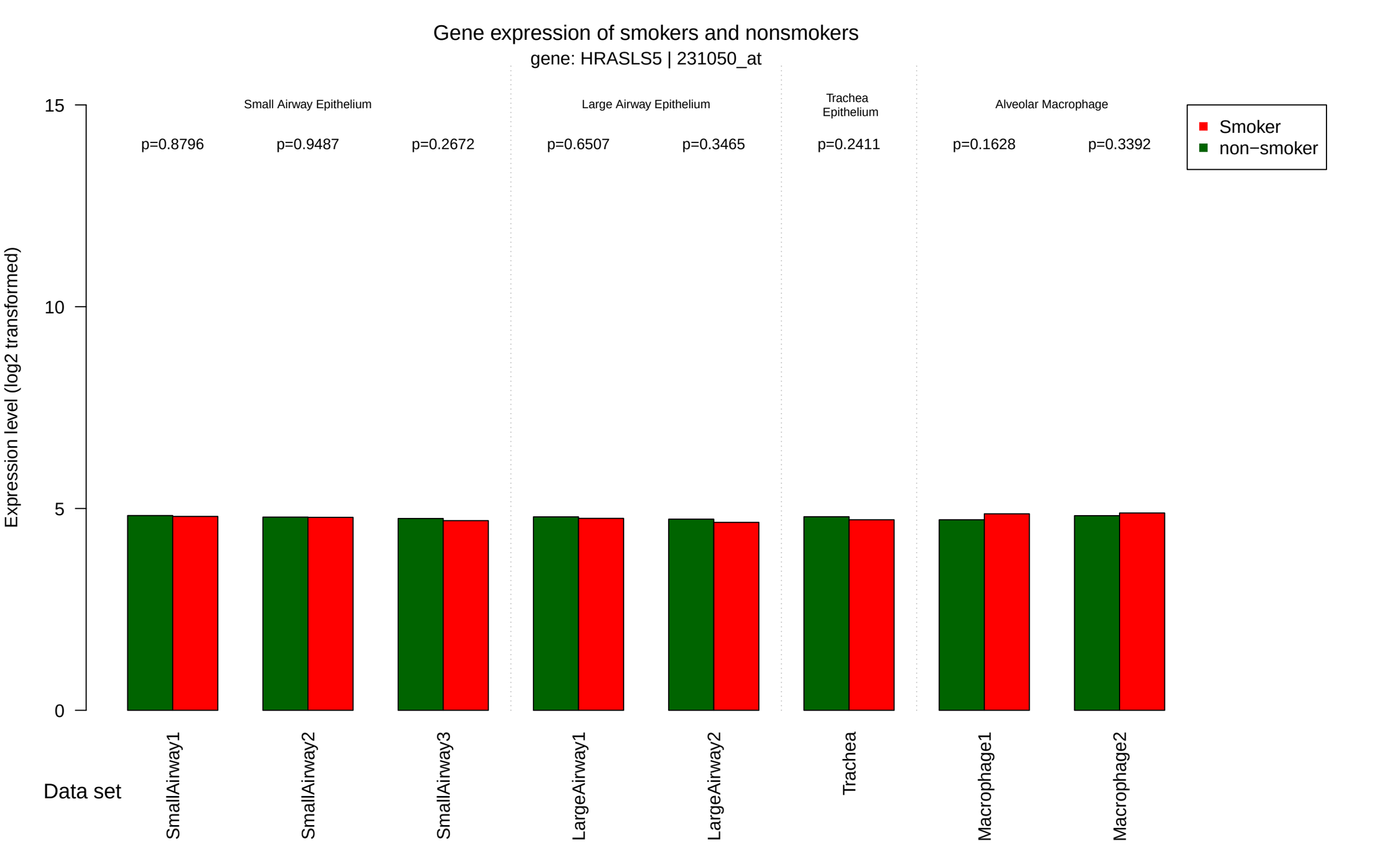
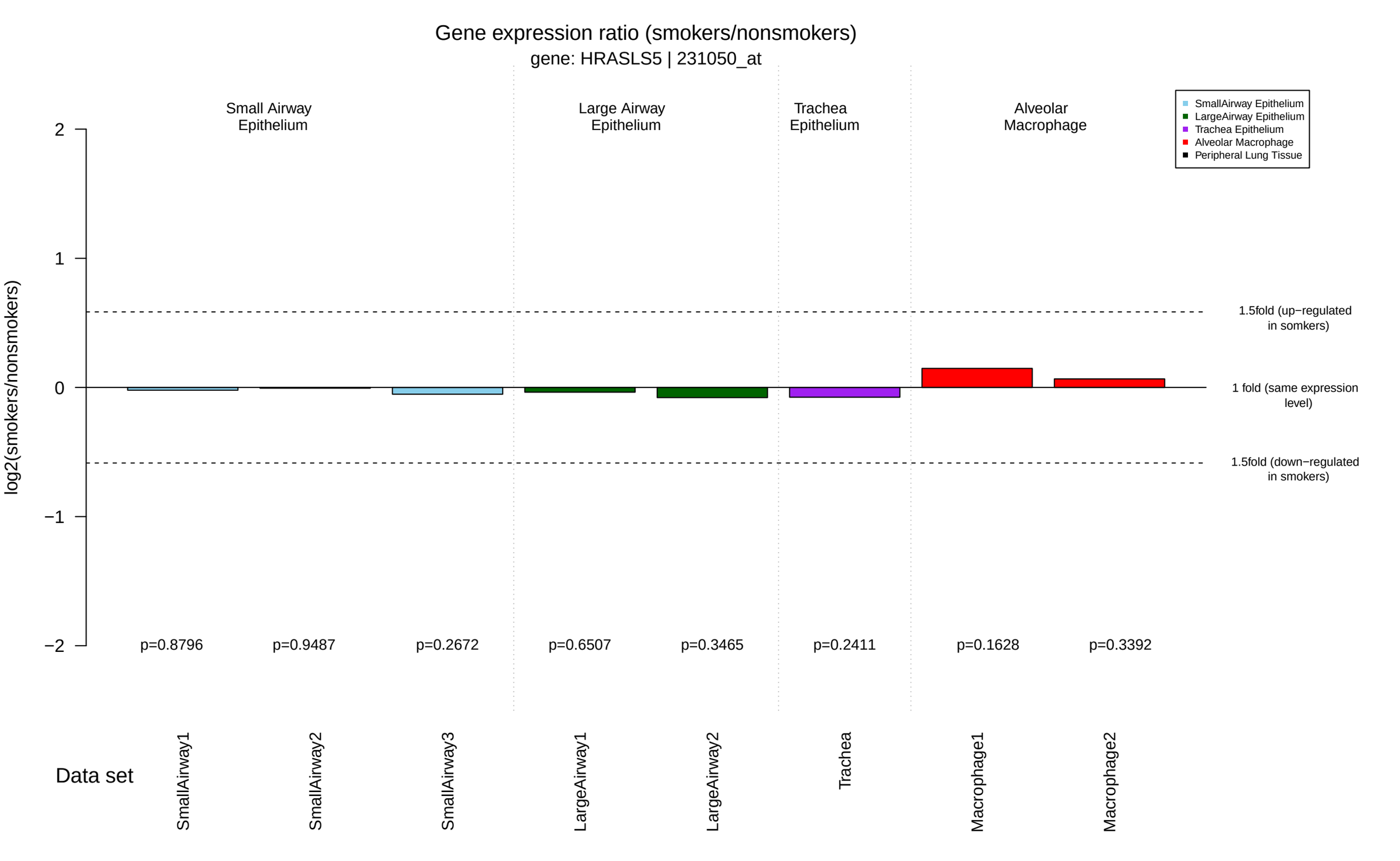
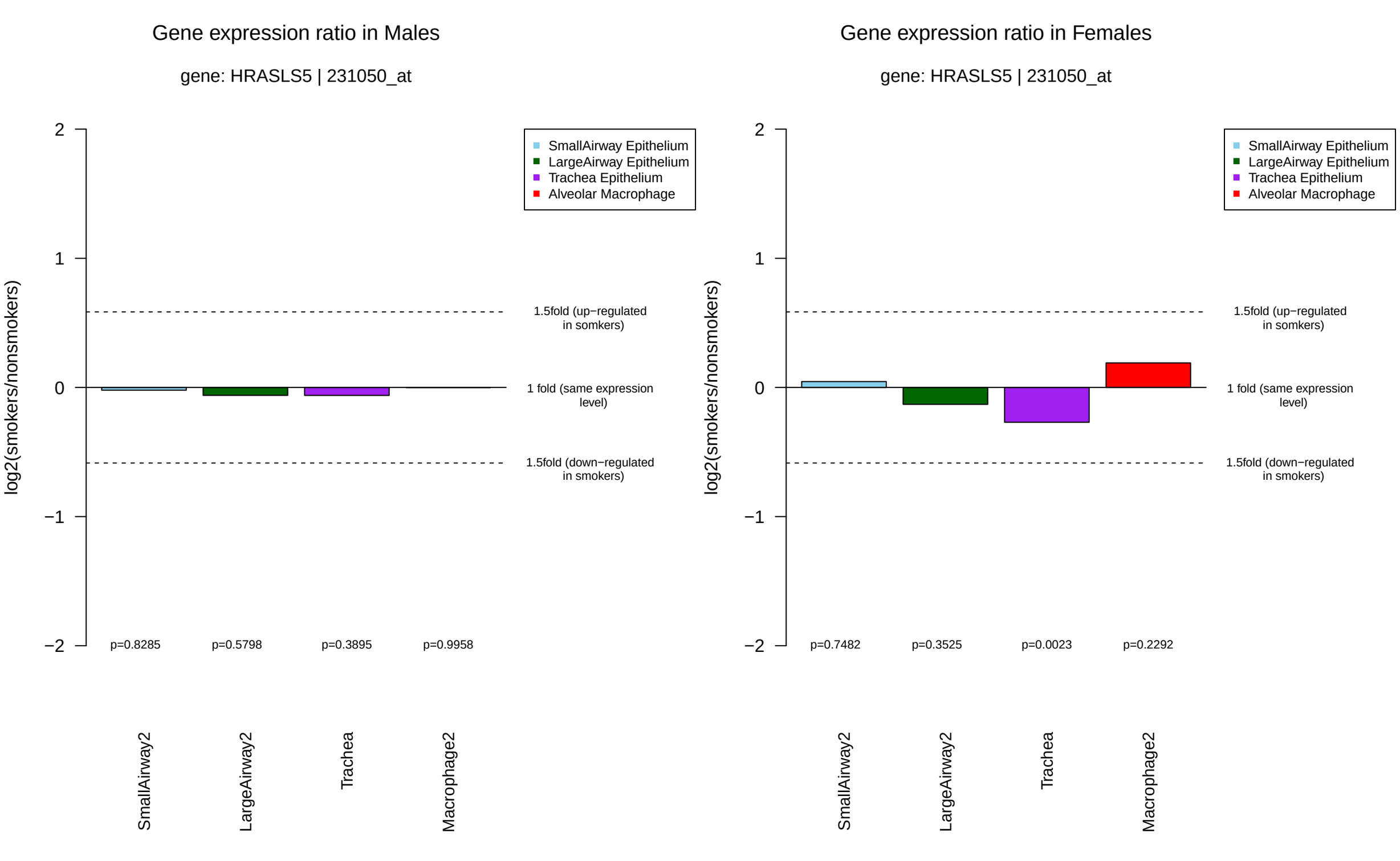
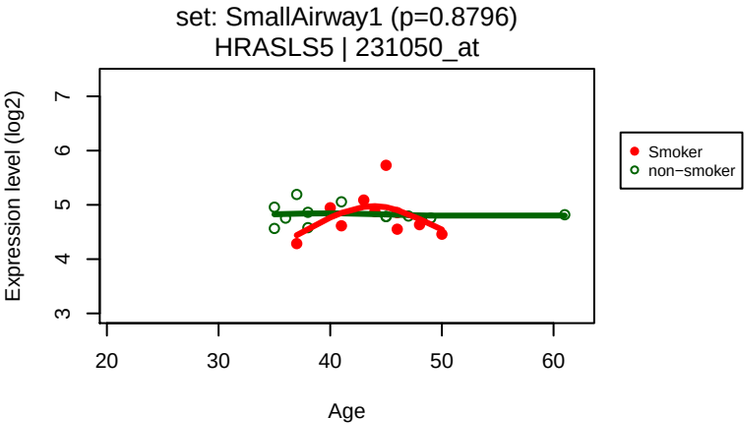
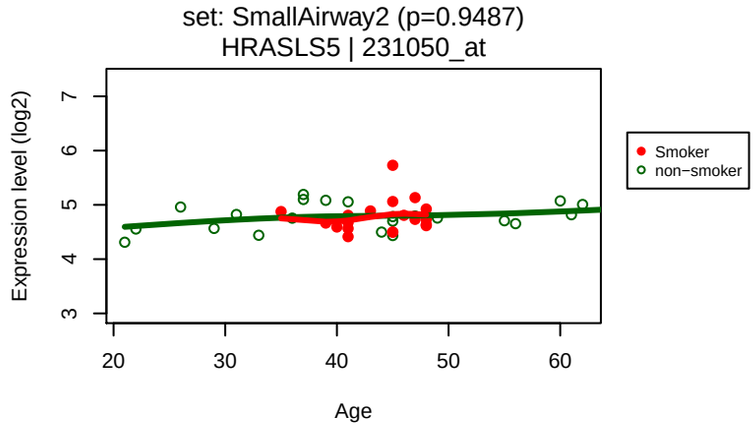
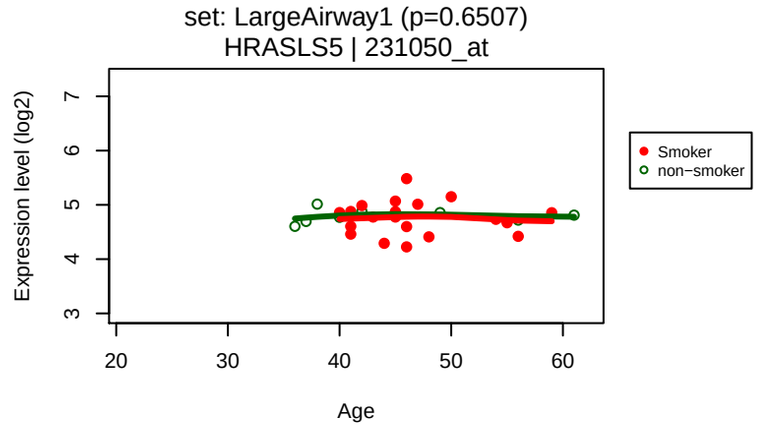

Gene expression of smokers and nonsmokers
gene: HRASLS5 | 231050_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8796
p=0.9487
p=0.2672
p=0.6507
p=0.3465
p=0.2411
p=0.1628
p=0.3392
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

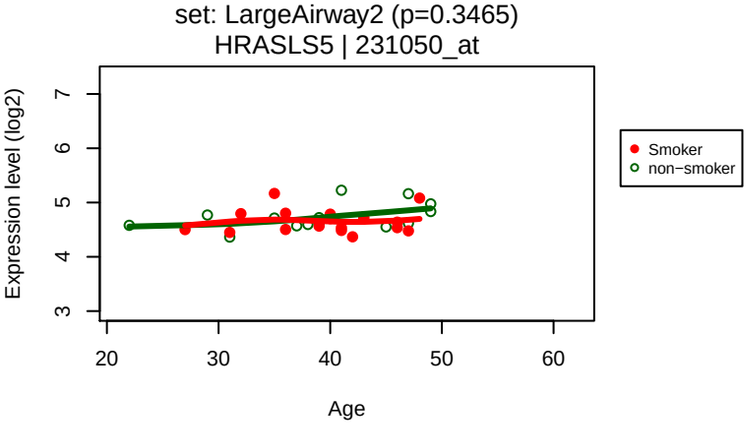

Gene expression ratio (smokers/nonsmokers)
gene: HRASLS5 | 231050_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8796
p=0.9487
p=0.2672
p=0.6507
p=0.3465
p=0.2411
p=0.1628
p=0.3392
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

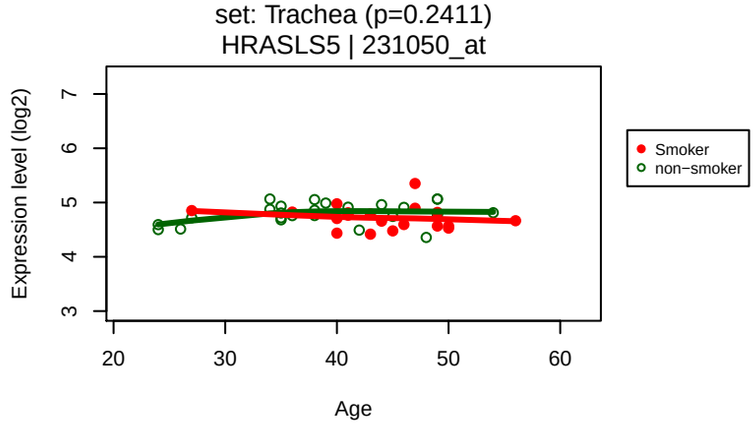

Gene expression ratio in Males
gene: HRASLS5 | 231050_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8285
p=0.5798
p=0.3895
p=0.9958
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HRASLS5 | 231050_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7482
p=0.3525
p=0.0023
p=0.2292
SmallAirway2
LargeAirway2
Trachea
Macrophage2

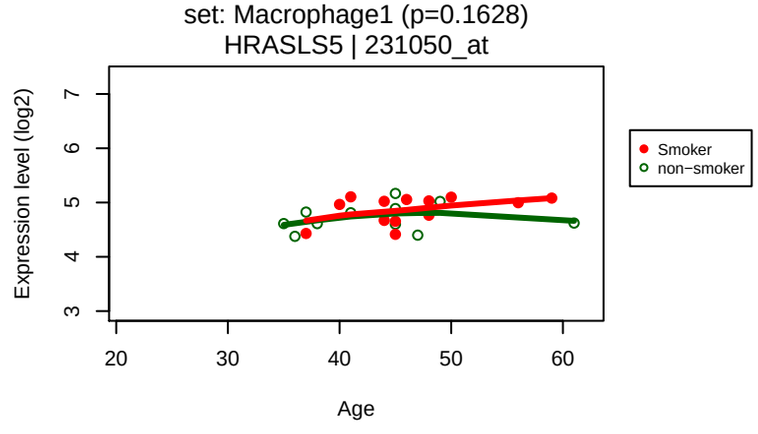

set: SmallAirway1 (p=0.8796)
HRASLS5 | 231050_at
Expression level (log2)
Age
Smoker
non−smoker

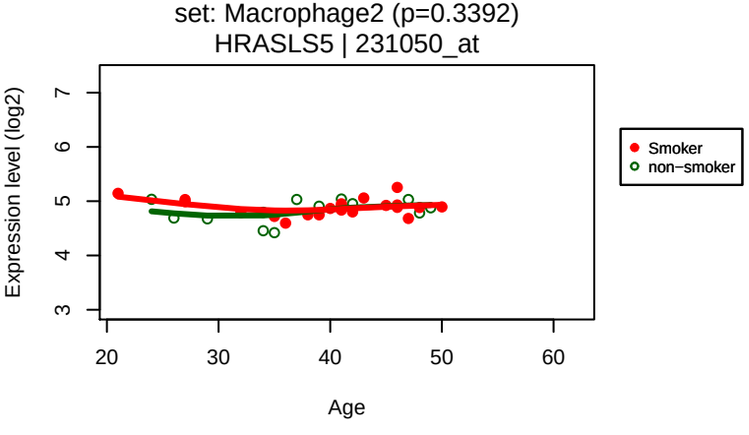

set: SmallAirway2 (p=0.9487)
HRASLS5 | 231050_at
Expression level (log2)
Age
Smoker
non−smoker

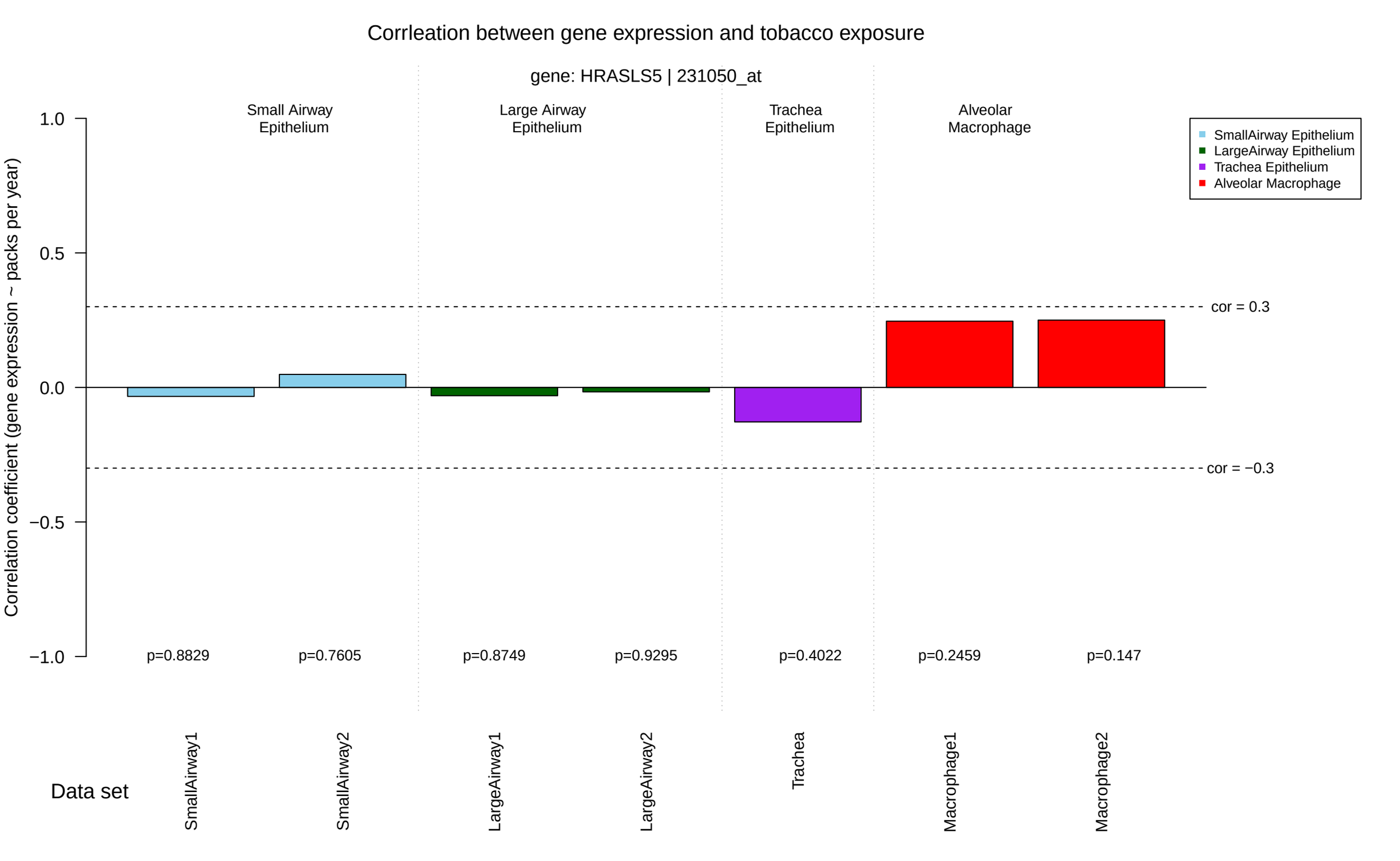

set: LargeAirway1 (p=0.6507)
HRASLS5 | 231050_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3465)
HRASLS5 | 231050_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2411)
HRASLS5 | 231050_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1628)
HRASLS5 | 231050_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3392)
HRASLS5 | 231050_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HRASLS5 | 231050_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8829
p=0.7605
p=0.8749
p=0.9295
p=0.4022
p=0.2459
p=0.147
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2